

SHANGLIN WU

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👤 shanglinwu.github.io

🎓 [Google Scholar](#)

Research Interests

My research focuses on understanding the fundamental principles that govern multi-agent AI systems, particularly how autonomous agents collaborate, learn, and adapt through interaction. I am especially interested in how collective behaviors emerge from individual agents, how agents learn continually from experience, and how coordination failures and emergent risks arise as these systems scale. My goal is to develop a deeper understanding of the mechanisms underlying learning, collaboration, and safety in increasingly autonomous multi-agent systems.

Education

Emory University, Computer Science

August 2025 – Now

Ph.D. in Computer Science and Informatics

Atlanta, USA

Advisor: Dr. Kai Shu

Peking University, Tong Class

September 2021 – May 2025

Bachelor of Artificial Intelligence, Computer Science

Beijing, China

Conferences

Shanglin Wu and Kai Shu. “Memory in LLM-based Multi-agent Systems: Mechanisms, Challenges, and Collective Intelligence.” *Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)*, 2026. **Best Survey Paper Award.**

Journal Papers

Wei-Chieh Huang, Weizhi Zhang, Yueqing Liang, ..., **Shanglin Wu**, ..., Kai Shu. “Rethinking Memory Mechanisms of Foundation Agents in the Second Half: A Survey.” *Transactions on Machine Learning Research (TMLR)*. *Core contributor.*

Preprints

Shanglin Wu, Yuyang Luo, Yueqing Liang, Kaiwen Shi, Yanfang Ye, Ali Payani, and Kai Shu. “Scaling Teams or Scaling Time? Memory Enabled Lifelong Learning in LLM Multi-Agent Systems” *arXiv preprint*, 2026.

Shanglin Wu, Lihui Liu, Jinho D. Choi, and Kai Shu. “Improving Factuality in LLMs via Inference-Time Knowledge Graph Construction.” *arXiv preprint*, 2025.

Research Experience

Research Intern

May 2026 - August 2026

Cisco Research

- Mentor: Ali Payani
- Developed WE-INTEND, a multi-agent RL framework that trains LLM agents to coordinate via verifiable joint-intention primitives, improving collaboration across cooperative benchmarks without inference-time scaffolding.

Research Intern

March 2025 - July 2025

Microsoft Research Asia Alumni

- Mentor: Rui Ding
- Developed an LLM agent for conditional independence testing using SFT and RL. The agent achieves outstanding results across both CI-test tasks and broader causal discovery applications.

Teaching Experience

CS 470: Data Mining

Spring 2026

Graduate Teaching Assistant

Emory University

CS 370: Computer Science Practicum

Fall 2025

Graduate Teaching Assistant

Emory University

Awards

Excellence in Teaching Assistance Commendation

May 2026

Department of Computer Science, Emory University

Research Competition Paper Award

December 2023

The National Conference on Supply Chain and Operation Management(ISCOM) 2023

Scholarship

December 2021

Peking University Freshman Scholarship